Advanced Models

TBD

Load packages

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(dplyr)
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```r
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```r
# Including my traditional diagnostics package...  See the following for details:
# https://statacumen.com/teach/ADA1/help/Erhardt_R_Install.html
# The usual function call here is e_plot_lm_diagnostics(df, sw_plot_set = "simple")
if ((requireNamespace("erikmisc", quietly = TRUE)) == FALSE) {
  remotes::install_github("erikerhardt/erikmisc")
}
```

```
Warning: replacing previous import 'car::recode' by 'dplyr::recode' when
loading 'erikmisc'

Registered S3 methods overwritten by 'ggpp':
  method                  from
  heightDetails.titleGrob ggplot2
  widthDetails.titleGrob  ggplot2

Warning: replacing previous import 'ggplot2::stat_qq_line' by
'qqplotr::stat_qq_line' when loading 'erikmisc'

Warning: replacing previous import 'ggplot2::StatQqLine' by
'qqplotr::StatQqLine' when loading 'erikmisc'

Warning: replacing previous import 'dplyr::lag' by 'stats::lag' when loading
'erikmisc'

Warning: replacing previous import 'dplyr::filter' by 'stats::filter' when
loading 'erikmisc'

Warning: replacing previous import 'stats::filter' by 'dplyr::filter' when
loading 'erikmisc'
```

```
Warning: replacing previous import 'car::ellipse' by 'ellipse::ellipse' when
loading 'erikmisc'
```

```
Warning: replacing previous import 'ggplot2::annotate' by 'ggpp::annotate' when
loading 'erikmisc'
```

```
library(erikmisc)

# Set seed for reproducability.
set.seed(6979)
```

Load the Data

```
#Load the data
ed_data <- read.csv("final_data.csv")

#Remove entries without median debt reported, and
# turn it into numerical data.
ed_data <- ed_data |> subset(DEBT_MDN != "PS")
ed_data$DEBT_MDN <- as.numeric(ed_data$DEBT_MDN)

# Clean up all factors which were imported as integers.
ed_data$CONTROL.x <- as.factor(ed_data$CONTROL.x)
ed_data$REGION <- as.factor(ed_data$REGION)
ed_data$HIGHDEG <-as.factor(ed_data$HIGHDEG)
ed_data$PREDDEG <- as.factor(ed_data$PREDDEG)
ed_data$NUMBRANCH <- as.factor(ed_data$NUMBRANCH)
ed_data$MAIN.x <-as.factor(ed_data$MAIN.x)
ed_data$SCH_DEG <- as.factor(ed_data$SCH_DEG)

# Turn all numeric data that got read as strings
# back into numerical data, changing NA values to
# zero for simplicity.
cols_to_convert <- c(
  "DEATH_YR8_RT",
  "COMP_ORIG_YR8_RT",
  "COMP_4YR_TRANS_YR8_RT",
  "PELL_COMP_4YR_TRANS_YR8_RT",
  "PELL_WDRAW_ORIG_YR8_RT",
  "NOPELL_WDRAW_ORIG_YR8_RT",
```

```
  "NOT1STGEN_COMP_4YR_TRANS_YR2_RT",
  "FIRSTGEN_COMP_ORIG_YR2_RT",
  "DEBT_MDN",
  "CREDDESC",
  "DEBT_ALL_STGP_ANY_MEAN",
  "BBRR1_FED_UG_DFLT"
)

new_ed_data <- ed_data %>%
  mutate(across(all_of(cols_to_convert), ~ {
    x <- suppressWarnings(as.numeric(.x))
    replace(x, is.na(x), 0)
  }))

# Now center our data. Since our original data had many character columns that should have be
centered_ed_data <- new_ed_data
numeric_cols <- sapply(ed_data, is.numeric)
centered_ed_data[numeric_cols] <- lapply(centered_ed_data[numeric_cols],
                                         scale, scale = FALSE)
rm(numeric_cols)
```

```
model_1 <- lm(EARN_MDN_4YR ~ .-CITY, data = centered_ed_data)
model2 <- stepAIC(model_1, direction="forward")
```

```
Start:  AIC=616680.9
EARN_MDN_4YR ~ (X + OPEID6 + CITY + SCH_DEG + MAIN.x + NUMBRANCH +
    PREDDEG + HIGHDEG + ST_FIPS + REGION + ADM_RATE_ALL + PCIP12 +
    PCIP24 + PCIP51 + PCIP52 + UGDS + UG25ABV + UGDS_WHITE +
    UGDS_BLACK + UGDS_HISP + PPTUG_EF + COSTT4_A + TUITIONFEE_IN +
    TUITIONFEE_OUT + TUITFTE + INEXPFTE + AVGFACSAL + PFTFAC +
    C150_4 + PFTFTUG1_EF + D150_4 + C150_4_WHITE + C150_4_BLACK +
    C150_4_HISP + DEATH_YR8_RT + COMP_ORIG_YR8_RT + COMP_4YR_TRANS_YR8_RT +
    PELL_COMP_4YR_TRANS_YR8_RT + PELL_WDRAW_ORIG_YR8_RT + NOPELL_WDRAW_ORIG_YR8_RT +
    NOT1STGEN_COMP_4YR_TRANS_YR2_RT + FIRSTGEN_COMP_ORIG_YR2_RT +
    TUITFTE.1 + OVERALL_YR2_N + DEBT_MDN + CREDDESC + DEBT_ALL_STGP_ANY_MEAN +
    BBRR1_FED_UG_DFLT + DISTANCE + CONTROL.x + CONTROL.y + SATVRMID +
    SATMTMID) - CITY
```

```
model_3 <- lm(
  EARN_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2
  + PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN,
```

```
  data = centered_ed_data
)
model_3_unshifted <- model_3 <- lm(
  EARN_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2
  + PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN,
  data = ed_data
)
```

1. Multiple regression models (10 pts)

At least two candidate models with several predictors are fit. Coefficients are interpreted correctly, holding the other predictors fixed. Categorical predictors are coded and interpreted correctly.

Model 1

Model 2

Model 3

For our third, model, we took a more hand-built approach. We included in-state tuition fee, institution type (public, private non-profit, or private for-profit), and admission rate, as well as all degree two interaction terms between these; predicted degree accomplishment, highest degree achieved, completion rate, median debt, and the interaction term between completion rate and institution admition rate. This gives us the following:

$$Y = ...$$

Here, we interpret...

2. Interactions and polynomial terms (10 pts)

At least one interaction or polynomial term is considered, with a reason for it. It is tested (for example, a partial F test or a test) and the result is stated. An interaction plot or fitted-curve plot is included where relevant. A well-reasoned term that turns out not to help still earns full credit.

Model 1

Model 2

Model 3

Here, we include degree two interaction terms between in-state tuition, institution type, and admission rate. This is because the effect of in-state tuition can be mitigated by private institutions waiving tuition, in-state tuition can often be waived by higher admittance requirements (reflected by admission rate), and institution type and admission rate together can tell us how selective the institution is, often indicating how well perceived the institution is in the job market.

There is also room for C150_4 to have interaction terms with practically every other variable present, as drop-out rate is a very useful indicator of median pay after graduation. Given the nature of graduation rate and admittance criteria, they will likely have interaction in that the more strict the admittance criteria, the higher the graduation rate will likely be. As such, we try to account for how this effects pay by including an interaction term.

The remaining items: median debt, predicted degree, and highest degree received, all indicate base-line requirements for most students at these institutions, and help us account for what the average student at these institutions look like.

3. Diagnostics (10 pts)

Residual plots and checks for the model's assumptions are shown. Outliers, influential points, or multicollinearity (VIF) are examined. The report says what the checks found, even if nothing was wrong.

Model 1

Model 2

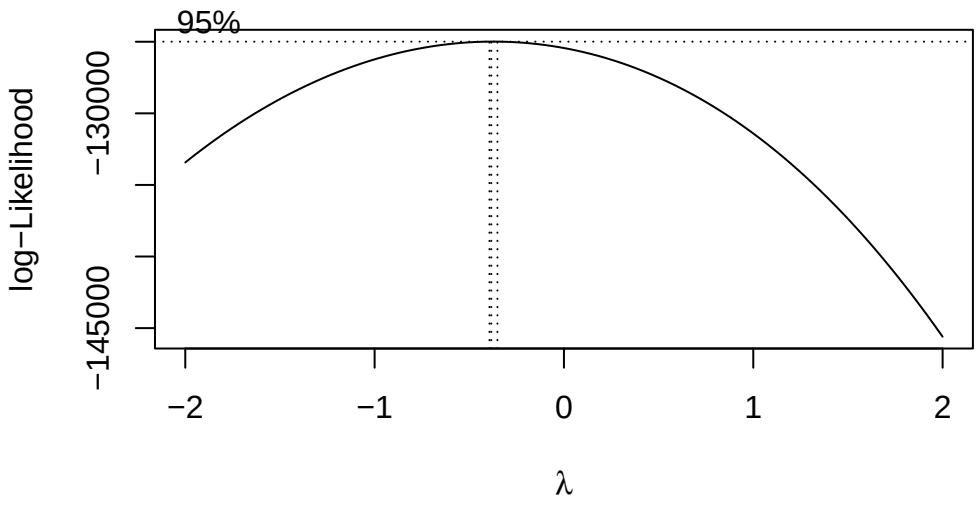
Model 3

```
#e_plot_lm_diagnostics(model_3, sw_plot_set = "full")
vif(model_3)
```

```
there are higher-order terms (interactions) in this model
consider setting type = 'predictor'; see ?vif
```

```
                                GVIF Df GVIF^(1/(2*Df))
TUITIONFEE_IN                124.201020  1       11.144551
CONTROL.x                  18684.930411  2       11.691571
ADM_RATE_ALL                  23.279770  1        4.824911
PREDDEG                        1.045150  1        1.022326
HIGHDEG                        1.126859  2        1.030309
C150_4                        20.590276  1        4.537651
DEBT_MDN                       1.739878  1        1.319045
TUITIONFEE_IN:CONTROL.x     1343.570214  2        6.054316
TUITIONFEE_IN:ADM_RATE_ALL    70.150226  1        8.375573
CONTROL.x:ADM_RATE_ALL     12280.261710  2       10.526934
ADM_RATE_ALL:C150_4           31.355582  1        5.599606
```

```
bc3 <- boxcox(model_3_unshifted)
```




```
bc3
```

```
$x
  [1] -2.00000000 -1.95959596 -1.91919192 -1.87878788 -1.83838384 -1.79797980
  [7] -1.75757576 -1.71717172 -1.67676768 -1.63636364 -1.59595960 -1.55555556
 [13] -1.51515152 -1.47474747 -1.43434343 -1.39393939 -1.35353535 -1.31313131
 [19] -1.27272727 -1.23232323 -1.19191919 -1.15151515 -1.11111111 -1.07070707
```

```
 [25] -1.03030303 -0.98989899 -0.94949495 -0.90909091 -0.86868687 -0.82828283
 [31] -0.78787879 -0.74747475 -0.70707071 -0.66666667 -0.62626263 -0.58585859
 [37] -0.54545455 -0.50505051 -0.46464646 -0.42424242 -0.38383838 -0.34343434
 [43] -0.30303030 -0.26262626 -0.22222222 -0.18181818 -0.14141414 -0.10101010
 [49] -0.06060606 -0.02020202  0.02020202  0.06060606  0.10101010  0.14141414
 [55]  0.18181818  0.22222222  0.26262626  0.30303030  0.34343434  0.38383838
 [61]  0.42424242  0.46464646  0.50505051  0.54545455  0.58585859  0.62626263
 [67]  0.66666667  0.70707071  0.74747475  0.78787879  0.82828283  0.86868687
 [73]  0.90909091  0.94949495  0.98989899  1.03030303  1.07070707  1.11111111
 [79]  1.15151515  1.19191919  1.23232323  1.27272727  1.31313131  1.35353535
 [85]  1.39393939  1.43434343  1.47474747  1.51515152  1.55555556  1.59595960
 [91]  1.63636364  1.67676768  1.71717172  1.75757576  1.79797980  1.83838384
 [97]  1.87878788  1.91919192  1.95959596  2.00000000

$y
  [1] -133428.2 -133006.7 -132596.6 -132198.0 -131810.7 -131434.6 -131069.7
  [8] -130716.0 -130373.3 -130041.6 -129720.9 -129411.1 -129112.1 -128823.9
 [15] -128546.4 -128279.7 -128023.5 -127777.9 -127542.9 -127318.4 -127104.3
 [22] -126900.7 -126707.5 -126524.7 -126352.2 -126190.0 -126038.1 -125896.5
 [29] -125765.2 -125644.1 -125533.3 -125432.7 -125342.4 -125262.3 -125192.5
 [36] -125132.9 -125083.6 -125044.6 -125015.9 -124997.5 -124989.5 -124991.9
 [43] -125004.7 -125028.0 -125061.8 -125106.1 -125161.0 -125226.5 -125302.8
 [50] -125389.8 -125487.7 -125596.4 -125716.1 -125846.8 -125988.7 -126141.7
 [57] -126306.0 -126481.7 -126668.8 -126867.5 -127077.8 -127299.9 -127533.9
 [64] -127779.8 -128037.7 -128307.9 -128590.4 -128885.2 -129192.7 -129512.7
 [71] -129845.6 -130191.4 -130550.2 -130922.1 -131307.4 -131706.0 -132118.2
 [78] -132544.1 -132983.8 -133437.4 -133905.0 -134386.9 -134883.0 -135393.5
 [85] -135918.6 -136458.3 -137012.7 -137582.0 -138166.2 -138765.5 -139379.8
 [92] -140009.4 -140654.3 -141314.5 -141990.1 -142681.1 -143387.7 -144109.7
 [99] -144847.3 -145600.5
```

```
bc3$x[which.max(bc3$y)]
```

```
[1] -0.3838384
```

4. Model selection and justification (10 pts)

Candidate models are compared with a stated criterion (adjusted R^2, AIC, BIC, C_p, or an F test). The final model is chosen and the choice is justified in words. Note: STAT 540 groups: deeper selection justification In addition to the above, a 540 group compares at least three candidate models using at least two different criteria. Where the criteria disagree, the report

explains why and says what the group traded off (fit against parsimony, or prediction against interpretation). Award 10 only if this is done. This is part of the criterion, not extra points.

Model 1

Model 2

Model 3

5. Integration and reproducibility (10 pts) - COMPLETE

The report builds on Milestone 2 and responds to its feedback. The submission is a rendered PDF named Group_XX_Milestone3.pdf, with code visible, a seed set, and sessionInfo() present.

For Diagnostics and Reproducibility:

```
sessionInfo()
```

```
R version 4.4.3 (2025-02-28 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default


locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/Denver
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
```

```
 [1] erikmisc_0.2.91 MASS_7.3-65     car_3.1-3       carData_3.0-5
 [5] lubridate_1.9.4 forcats_1.0.1   stringr_1.6.0   dplyr_1.1.4
 [9] purrr_1.2.1     readr_2.1.6     tidyr_1.3.2     tibble_3.3.1
[13] ggplot2_4.0.1   tidyverse_2.0.0

loaded via a namespace (and not attached):
  [1] fs_2.0.1                bitops_1.0-9            insight_1.6.0
  [4] RColorBrewer_1.1-3      doParallel_1.0.17       numDeriv_2016.8-1.1
  [7] tools_4.4.3             backports_1.5.0         sjlabelled_1.2.0
 [10] R6_2.6.1                mgcv_1.9-4              nortest_1.0-4
 [13] jomo_2.7-6              withr_3.0.2             gridExtra_2.3
 [16] quantreg_6.1            cli_3.6.5               textshaping_1.0.4
 [19] officer_0.7.7           sandwich_3.1-1          cAIC4_1.1
 [22] mvtnorm_1.3-3           S7_0.2.1                robustbase_0.99-7
 [25] askpass_1.2.1           systemfonts_1.3.1       yulab.utils_0.2.5
 [28] foreign_0.8-90          svglite_2.2.2           qqplotr_0.0.7
 [31] DHARMa_0.5.0            parallelly_1.46.1       labelled_2.16.0
 [34] readxl_1.6.0            pwr_1.3-0               rstudioapi_0.18.0
 [37] visNetwork_2.1.4        generics_0.1.4          gridGraphics_0.5-1
 [40] shape_1.4.6.1           sjPlot_2.9.0            zip_2.3.3
 [43] leaps_3.2               Matrix_1.7-4            ggbeeswarm_0.7.3
 [46] abind_1.4-8             lifecycle_1.0.5         multcomp_1.4-29
 [49] yaml_2.3.12             gvlma_1.0.0.3           snakecase_0.11.1
 [52] ggpmisc_1.0.0           recipes_1.3.1           grid_4.4.3
 [55] crayon_1.5.3            mitml_0.4-5             lattice_0.22-7
 [58] haven_2.5.5             cowplot_1.2.0           pillar_1.11.1
 [61] knitr_1.51              log4r_0.5.0             boot_1.3-32
 [64] estimability_1.5.1      future.apply_1.20.1     codetools_0.2-20
 [67] pan_2.0                 glue_1.8.0              fontLiberation_0.1.0
 [70] data.table_1.18.0       vctrs_0.7.0             png_0.1-9
 [73] Rdpack_2.6.4            cellranger_1.1.0        gtable_0.3.6
 [76] twosamples_2.0.1        ggpp_0.6.1              gower_1.0.2
 [79] xfun_0.56               rbibutils_2.4           prodlim_2025.04.28
 [82] pracma_2.4.6            coda_0.19-4.1           reformulas_0.4.3.1
 [85] survival_3.8-6          randomForestSRC_3.9.0   timeDate_4051.111
 [88] iterators_1.0.14        pbmcapply_1.5.1         tinytex_0.58
 [91] finalfit_1.1.0          hardhat_1.4.2           lava_1.8.2
 [94] DiagrammeR_1.0.12       TH.data_1.1-5           ROCR_1.0-12
 [97] ipred_0.9-15            nlme_3.1-168            tsne_0.2-0
[100] fontquiver_0.2.1        data.tree_1.2.0         vipor_0.4.7
[103] otel_0.2.0              rpart_4.1.24            colorspace_2.1-2
[106] Hmisc_5.3-0             nnet_7.3-20             tidyselect_1.2.1
[109] emmeans_2.0.1           compiler_4.4.3          glmnet_4.1-10
```

```
[112] htmlTable_2.5.0        flextable_0.10.3       mice_3.19.0
[115] SparseM_1.84-2         xml2_1.5.2             fontBitstreamVera_0.1.1
[118] checkmate_2.3.4        scales_1.4.0           caTools_1.18.3
[121] DEoptimR_1.1-4         rappdirs_0.3.4         digest_0.6.39
[124] minqa_1.2.8            rmarkdown_2.30         htmltools_0.5.9
[127] pkgconfig_2.0.3        base64enc_0.1-3        lme4_1.1-38
[130] umap_0.2.10.0          fastmap_1.2.0          rlang_1.1.7
[133] htmlwidgets_1.6.4      RLRsim_3.1-9           farver_2.1.2
[136] zoo_1.8-15             jsonlite_2.0.0         ModelMetrics_1.2.2.2
[139] magrittr_2.0.4         polynom_1.4-1          kableExtra_1.4.0
[142] Formula_1.2-5          ggplotify_0.1.4        patchwork_1.3.2
[145] Rcpp_1.1.1             gdtools_0.5.2          reticulate_1.45.0
[148] stringi_1.8.7          pROC_1.19.0.1          plyr_1.8.9
[151] parallel_4.4.3         listenv_0.10.0         sjmisc_2.8.11
[154] splines_4.4.3          hms_1.1.4              qqconf_1.3.2
[157] ggpubr_0.6.2           uuid_1.2-1             opdisDownsampling_1.7
[160] ggsignif_0.6.4         reshape2_1.4.5         stats4_4.4.3
[163] evaluate_1.0.5         modelr_0.1.11          nloptr_2.2.1
[166] tzdb_0.5.0             foreach_1.5.2          moonBook_0.3.1
[169] MatrixModels_0.5-4     openssl_2.3.4          future_1.69.0
[172] janitor_2.2.1          broom_1.0.11           xtable_1.8-4
[175] RSpectra_0.16-2        rstatix_0.7.3          viridisLite_0.4.2
[178] class_7.3-23           ragg_1.5.0             lmerTest_3.2-0
[181] beeswarm_0.4.0         ellipse_0.5.0          cluster_2.1.8.1
[184] corrplot_0.95          timechange_0.3.0       globals_0.18.0
[187] caret_7.0-1
```